FINAL EXAMINATION — MOCK TEST PAPER (Syllabus 2022)

PAPER 15: STRATEGIC COST MANAGEMENT

Set for the December 2026 term · Pattern as per the December 2025 paper

Time Allowed: 3 Hours | Full Marks: 100

The figures in the margin on the right side indicate full marks.
This paper contains two sections, A and B. **Section A is compulsory** (15 × 2 = 30 marks). In **Section B**, answer any **five** questions out of seven (Q.2 to Q.8), 14 marks each (5 × 14 = 70 marks). Working notes should form part of the answer. Wherever necessary, make suitable assumptions and state them clearly. **PYQ Dec 2022** etc. shows the exam in which the question (or one of the same type) was asked, as tagged in the SCM Scanner.

SECTION – A (15 × 2 = 30 Marks)

1. Choose the correct answer from the four alternatives given. (Give the justification / working wherever applicable.)

(i) Pareto analysis, used in quality management, rests on the principle that: **PYQ Dec 2022**

A. 20% of the causes give rise to 20% of the problems
B. 80% of the problems arise from 20% of the causes
C. All causes have equal weight
D. 50% of the problems arise from 50% of the causes

(ii) Which of the following is an *appraisal cost* of quality? **PYQ Dec 2017** **PYQ Jun 2025**

A. Training of quality-control staff
B. Rework of defective units
C. Inspection of incoming raw materials
D. Warranty repairs

(iii) In Lean Accounting, costs are normally collected and reported by: **PYQ Jun 2024**

A. Cost centres
B. Value streams
C. Individual products using full absorption
D. Departments

(iv) Sales are ₹10,00,000, break-even sales are ₹6,00,000 and the P/V ratio is 40%. The profit is: **PYQ Dec 2019**

A. ₹4,00,000
B. ₹2,40,000
C. ₹1,60,000
D. ₹1,00,000

(v) The first unit of a product takes 100 hours. With an 80% learning curve, the total time for the first 4 units will be: **PYQ Jun 2016**

A. 320 hours
B. 256 hours
C. 400 hours
D. 204.8 hours

(vi) A firm's demand function is $p = 120 - 2x$ and its total cost function is $C = x^2 + 24x + 40$. The output that maximises profit is: **PYQ Jun 2016** **PYQ Dec 2021**

A. 12 units
B. 16 units
C. 20 units
D. 24 units

(vii) In the Hungarian method for an $n \times n$ assignment problem, the solution is optimal when: **PYQ Dec 2021**

A. The minimum number of lines covering all zeros equals n
B. Every row has at least two zeros
C. The number of lines is less than n
D. All elements become zero

(viii) A transportation problem has 4 sources and 5 destinations. A basic feasible solution is non-degenerate only if the number of occupied cells is: **PYQ Jun 2025** **PYQ Dec 2018**

A. 9
B. 20
C. 8
D. 5

(ix) For a PERT activity, the optimistic time is 4 days, the most likely time is 6 days and the pessimistic time is 14 days. The expected time and the standard deviation are: **PYQ Dec 2021**

A. 8 days; 1.67
B. 7 days; 2.78
C. 7 days; 1.67
D. 6 days; 1.67

(x) In a simulation exercise using two-digit random numbers (00–99), the demand levels have probabilities 0.10, 0.25, 0.40 and 0.25. The random-number range for the second demand level is: **PYQ Dec 2021**

A. 00 – 24
B. 10 – 34
C. 10 – 35
D. 11 – 35

(xi) OLAP (On-Line Analytical Processing) is best described as: **PYQ Jun 2023** **PYQ Jun 2017**

A. Software for recording day-to-day transactions
B. Software that lets managers analyse data in many dimensions quickly
C. A tool for data entry validation

D. A network security protocol

(xii) Division X makes a component with a variable cost of ₹40, sells it outside at ₹60 and is working at full capacity. Selling costs of ₹5 per unit are saved on internal transfers. The minimum transfer price to Division Y is: **PYQ Jun 2025**

A. ₹40
B. ₹45
C. ₹55
D. ₹60

(xiii) A product now sells at ₹27,000. Competition will force a 15% price cut, and the company wants a margin of 20% on the selling price. The target cost per unit is: **PYQ Jun 2025**

A. ₹21,600
B. ₹18,360
C. ₹22,950
D. ₹19,125

(xiv) The standard rate is ₹50 per hour and the standard hours for actual output are 1,000. Actual hours worked are 1,100. The labour efficiency variance is: **PYQ Jun 2023**

A. ₹5,000 (F)
B. ₹5,500 (A)
C. ₹5,000 (A)
D. ₹500 (A)

(xv) Total cost of 10,000 units is ₹40,00,000 and capital employed is ₹20,00,000. Under rate-of-return pricing with a desired return of 15%, the selling price per unit is: **PYQ Dec 2021**

A. ₹460
B. ₹415
C. ₹400
D. ₹430

SECTION – B (5 × 14 = 70 Marks)

Answer any five questions out of seven questions given.

2. Decision Making

(a) A company made and sold 60,000 units last year. Its variable costs were ₹78,00,000 and its fixed costs ₹27,00,000. The normal selling price is ₹200 per unit. A foreign buyer offers to buy 5,000 units at ₹150 per unit. There is enough spare capacity to make the order without hurting normal sales. The order needs extra shipping of ₹83,000 and a one-time export licence fee of ₹40,000. **PYQ Dec 2021**
(i) Should the company accept the order? Show workings.
(ii) What is the minimum price per unit at which the order can be accepted? **5+3=8**

(b) "Fixed costs are irrelevant for decision making." Do you agree? Describe the situations in which fixed costs become relevant for decision making. **PYQ Dec 2022** **PYQ Dec 2019** **6**

3. Cost Management — Target Costing, Value Engineering and BPR

(a) Navya Ltd. sells a product at ₹27,000 per unit. Its present cost per unit is: direct material ₹9,000; direct labour ₹4,500; variable overhead ₹3,000; fixed overhead (allocated) ₹4,200. A competitor's entry is expected to force a price cut of 15%. The company wants a profit margin of 20% on the new selling price. A value-engineering team suggests: (1) redesigning the product to cut material cost by 8%; (2) partial automation to cut labour cost by 10%; (3) process improvement to cut variable overhead by 6%. **PYQ Jun 2025**

(i) Find the target cost and the cost gap.

(ii) Find how much of the cost gap the proposals close, and the percentage cut in fixed overhead still needed to reach the target cost. **3+5=8**

(b) State the principles of Business Process Re-engineering (BPR). **PYQ Jun 2023** **PYQ Dec 2017** **6**

4. Quality Management

(a) The following information for the year 2025-26 relates to Shubh Ltd., which sold 10,000 units at ₹2,000 each: **PYQ Jun 2023** **PYQ Dec 2021**

Particulars	₹
Training of quality-control staff	3,00,000
Design reviews	2,00,000
Inspection of incoming materials	4,50,000
Product testing (in-process and final)	3,50,000
Scrap and rework	6,00,000
Machine downtime due to defects	2,50,000
Warranty repairs	5,00,000
Sales allowances for sub-standard products	1,30,000

In addition, 200 units of sales were lost because of quality problems. The contribution is ₹800 per unit. Prepare a Cost of Quality report. Classify the costs into prevention, appraisal, internal failure and external failure costs, and express each category as a percentage of sales. **8**

(b) Summarise the essential steps in implementing Total Quality Management (TQM). **PYQ Jun 2023** **PYQ Jun 2015** **6**

5. Transfer Pricing and Pricing Strategy

(a) Division A of a company makes a component. It has a capacity of 50,000 units, an external demand of 40,000 units at ₹100 each, and a variable cost of ₹60 per unit. Division B needs 15,000 units of the component, which it can buy outside at ₹95 per unit. **PYQ Jun 2025** **PYQ Dec 2024**

(i) Find the minimum transfer price per unit that Division A should accept for all 15,000 units.

(ii) Recommend the best sourcing plan for the company as a whole, and quantify the benefit.

(iii) Give the range of transfer prices that would motivate both divisions to follow the plan in (ii). **3+4+1=8**

(b) Write a note on Price Sensitivity. Also mention the factors that affect it. **PYQ Dec 2019** **6**

6. Linear Programming and Learning Curve

(a) A firm makes two products, X and Y, with contributions of ₹30 and ₹40 per unit. Each unit of X needs 2 machine hours and 2 labour hours. Each unit of Y needs 3 machine hours and 1 labour hour. 120 machine hours and 80 labour hours are available. Formulate the problem as an LPP. Solve it by the graphical method to find the product mix that maximises contribution. **PYQ Jun 2024** **PYQ Dec 2022** **8**

(b) The first batch of a new product takes 1,000 labour hours at ₹20 per hour, and an 80% learning curve applies. Calculate: (i) the labour cost of an order of 8 batches; (ii) the extra labour hours needed if a repeat order of 8 more batches is received. **PYQ Jun 2016** **PYQ Dec 2015** **6**

7. Transportation and Assignment

(a) Find an initial solution by Vogel's Approximation Method and test it for optimality: **PYQ Jun 2025** **PYQ Dec 2018**

Source / Dest. (₹ per unit)	D1	D2	D3	Supply
S1	4	6	8	50
S2	5	4	7	60
S3	6	8	5	40
Demand	40	70	40	150

8

(b) A sales manager must assign four salesmen (A, B, C, D) to four territories (W, X, Y, Z). The cost (₹'000) of each assignment is given below. Find the assignment with the lowest total cost. **PYQ Dec 2022** **PYQ Dec 2021**

Salesman	W	X	Y	Z
A	10	12	19	11
B	5	10	7	8
C	12	14	13	11
D	8	15	11	9

6

8. Network Analysis and Business Intelligence

(a) A project has the following activities: **PYQ Dec 2021** **PYQ Dec 2019**

Activity	A	B	C	D	E	F	G
Predecessor	—	—	A	A	B, C	D, E	B
Duration (weeks)	4	6	5	7	3	4	8

Draw the network. Find the critical path and the project duration, and compute the total float of each activity. **8**

(b) Write short notes on any **two** of the following: **3×2=6**

(i) Business Intelligence (BI) software **PYQ Jun 2023**

(ii) Data warehouse **PYQ Jun 2014**

(iii) Lean Accounting **PYQ Dec 2021** **PYQ Jun 2019**

*** End of Question Paper ***

SUGGESTED ANSWERS

Section A

Q	Ans	Working / Reason
(i)	B	80:20 rule: about 80% of problems come from about 20% of causes.
(ii)	C	Inspection measures conformance, so it is an appraisal cost. Training is prevention, rework is internal failure, warranty is external failure.
(iii)	B	Lean accounting uses value-stream costing.
(iv)	C	MOS = 10,00,000 − 6,00,000 = 4,00,000; Profit = 4,00,000 × 40% = ₹1,60,000.
(v)	B	Cumulative average for 4 units = 100 × 0.8 × 0.8 = 64 hrs; total = 64 × 4 = 256 hrs.
(vi)	B	$\pi = (120x - 2x^2) - (x^2 + 24x + 40) = 96x - 3x^2 - 40$; $d\pi/dx = 96 - 6x = 0 \Rightarrow x = 16$ ($d^2\pi/dx^2 = -6 < 0$).
(vii)	A	Optimality test: lines = order of the matrix.
(viii)	C	m + n − 1 = 4 + 5 − 1 = 8.
(ix)	C	$t_e = (4 + 4\times6 + 14)/6 = 7$; $\sigma = (14 - 4)/6 = 1.67$.
(x)	B	Cumulative: 0.10 → 00–09; 0.35 → 10–34.
(xi)	B	OLAP supports fast, multi-dimensional analysis.
(xii)	C	Full capacity: Min TP = VC + lost contribution = 40 + (60 − 40) − 5 saved = ₹55.
(xiii)	B	New price = 27,000 × 85% = 22,950; Target cost = 22,950 × 80% = ₹18,360.
(xiv)	C	SR × (SH − AH) = 50 × (1,000 − 1,100) = ₹5,000 (A).
(xv)	D	Cost per unit = 400; required return = 20,00,000 × 15% / 10,000 = 30; Price = ₹430.

Section B

Q2 (a)

Variable cost per unit = 78,00,000 / 60,000 = ₹130. Fixed costs stay the same, so they are irrelevant.

Incremental analysis — 5,000 units	**₹**
Incremental revenue (5,000 × 150)	7,50,000
Less: Variable cost (5,000 × 130)	(6,50,000)
Contribution	1,00,000
Less: Shipping	(83,000)
Less: Export licence	(40,000)
Net incremental loss	**(23,000)**

(i) Reject the order at ₹150 (unless there are strategic benefits, such as entering a new market).
(ii) Minimum price = 130 + (83,000 + 40,000)/5,000 = 130 + 24.60 = **₹154.60 per unit**.

Q2 (b)

Not entirely. Fixed costs that stay the same under every option are irrelevant. Fixed costs become relevant when: (1) the decision itself changes them, e.g., extra supervision or rent for a special order, or a step-up in fixed costs when capacity is expanded; (2) they can be avoided, e.g., on closing down a segment or shutting a plant temporarily; (3) they are incurred specifically for the proposal, e.g., hiring a machine or a licence fee; (4) in make-or-buy decisions, where buying lets the firm avoid some fixed costs; (5) in long-run pricing and capacity decisions, where all costs must be recovered; (6) as opportunity costs of committed facilities that could be let out.

Q3 (a)

(i) New price = 27,000 × 0.85 = ₹22,950. Target cost = 22,950 × 0.80 = **₹18,360**.
Present cost = 9,000 + 4,500 + 3,000 + 4,200 = ₹20,700. **Cost gap = ₹2,340**.
(ii) Savings: material 9,000 × 8% = 720; labour 4,500 × 10% = 450; variable overhead 3,000 × 6% = 180.
Total = **₹1,350**.
Gap still open = 2,340 − 1,350 = ₹990. Cut needed in fixed overhead = 990 / 4,200 = **23.57%** (fixed overhead must fall to ₹3,210 per unit).

Q3 (b)

Principles of BPR (Hammer): (1) organise around outcomes, not tasks; (2) let those who use the output of a process perform the process; (3) build information processing into the real work that produces the information; (4) treat geographically dispersed resources as if they were centralised; (5) link parallel activities instead of integrating their results; (6) put the decision point where the work is performed, and build control into the process; (7) capture information once, at the source.

Q4 (a) — Cost of Quality Report (Sales ₹2,00,00,000)

Category	₹	% of sales
Prevention: Training 3,00,000 + Design review 2,00,000	5,00,000	2.50
Appraisal: Incoming inspection 4,50,000 + Testing 3,50,000	8,00,000	4.00
Internal failure: Scrap & rework 6,00,000 + Downtime 2,50,000	8,50,000	4.25
External failure: Warranty 5,00,000 + Allowances 1,30,000 + Lost contribution (200 × 800) 1,60,000	7,90,000	3.95
Total cost of quality	**29,40,000**	**14.70**

Failure costs (8.2% of sales) are higher than conformance costs (6.5% of sales). Spending more on prevention should cut total COQ.

Q4 (b)

> Steps in implementing TQM: (1) identify customers and customer groups, using a team approach such as multi-voting; (2) identify what customers expect; (3) identify the products or services that meet those expectations; (4) identify the processes that deliver them; (5) set up measures of performance and benchmarks; (6) find the root causes of problems, e.g., with Pareto analysis or fishbone diagrams; (7) improve processes continuously (Kaizen, PDCA); (8) train and empower employees, with top-management commitment throughout.

Q5 (a)

> Spare capacity of A = 50,000 − 40,000 = 10,000 units.
> (i) For 15,000 units: 10,000 units at VC ₹60 = 6,00,000. The other 5,000 units displace outside sales, so each costs VC 60 + lost contribution 40 = ₹100, giving 5,00,000. Minimum TP = 11,00,000 / 15,000 = **₹73.33 per unit**.
> (ii) Look at each block. The first 10,000 units cost ₹60 to make inside against ₹95 outside, so make them inside. The next 5,000 have an opportunity cost of ₹100 against ₹95 outside, so buy them outside.
> Cost to the company: transfer all 15,000 = ₹11,00,000; buy all 15,000 = ₹14,25,000; **best plan** (transfer 10,000, buy 5,000) = 6,00,000 + 4,75,000 = **₹10,75,000**. This saves ₹25,000 against transferring everything, and ₹3,50,000 against buying everything.
> (iii) For the 10,000 units transferred, any TP from **₹60 to ₹95** leaves both divisions no worse off.

Q5 (b)

> Price sensitivity is how much the price of a product affects buyers' purchasing behaviour. It is measured through price elasticity of demand: the percentage change in quantity divided by the percentage change in price. Factors: availability of substitutes; whether the product is a necessity or a luxury; share of the buyer's income spent on it; brand loyalty and differentiation; switching costs; whether the buyer is the payer (e.g., expense-account purchases); perceived quality ("price–quality effect"); the time period, since demand is more elastic in the long run; and how easily prices can be compared (internet).

Q6 (a)

> Max Z = 30x + 40y subject to 2x + 3y ≤ 120 (machine); 2x + y ≤ 80 (labour); x, y ≥ 0.
> Corner points: O(0,0) → 0; (40,0) → 1,200; (0,40) → 1,600. At the intersection, subtracting the constraints gives 2y = 40, so y = 20 and x = 30, and Z = 900 + 800 = 1,700.
> **Optimum: X = 30 units, Y = 20 units, maximum contribution ₹1,700.** (Both resources are fully used.)

Q6 (b)

Cumulative batches	Cum. avg hrs	Total hrs
1	1,000	1,000
2	800	1,600
4	640	2,560
8	512	4,096
16	409.6	6,553.6

(i) Labour cost of 8 batches = 4,096 × 20 = **₹81,920**.
(ii) Extra hours for 8 more batches = 6,553.6 − 4,096 = **2,457.6 hours** (₹49,152).

Q7 (a)

VAM: Step 1: the highest penalty is 2 (a tie between S1, D2 and D3). Pick S1, whose least-cost cell is S1D1 (₹4), and allocate 40 there. D1 is now satisfied. Step 2: the row penalties are S1 = 2, S2 = 3, S3 = 3. Pick S2 and allocate 60 to S2D2 (₹4). Step 3: S3 has a penalty of 3; allocate 40 to S3D3 (₹5). Step 4: allocate the remaining 10 to S1D2.
Allocations: S1D1 = 40, S1D2 = 10, S2D2 = 60, S3D3 = 40. Cost = 160 + 60 + 240 + 200 = **₹660**.
Degeneracy: there are 4 occupied cells, but m + n − 1 = 5. Put ε in independent cell S1D3.
MODI: $u_1 = 0 \rightarrow v_1 = 4$, $v_2 = 6$, $v_3 = 8$; $u_2 = -2$; $u_3 = -3$.
Opportunity costs $\Delta_{ij} = c_{ij} - (u_i + v_j)$: S2D1 = 3, S2D3 = 1, S3D1 = 5, S3D2 = 5. All are positive, so the VAM solution is **optimal and unique**, at a minimum cost of ₹660.

Q7 (b)

Row reduction (subtract each row minimum: 10, 5, 11, 8):
A: 0 2 9 1 · B: 0 5 2 3 · C: 1 3 2 0 · D: 0 7 3 1
Column reduction (subtract each column minimum: 0, 2, 2, 0):
A: 0 0 7 1 · B: 0 3 0 3 · C: 1 1 0 0 · D: 0 5 1 1
Four lines are needed to cover all zeros, so the solution is optimal. Assignments: D → W (only zero in row D), A → X, B → Y, C → Z.
Optimal: A–X (12), B–Y (7), C–Z (11), D–W (8). Minimum total cost = ₹38,000.

Q8 (a)

Activity	Dur.	ES	EF	LS	LF	Total float
A	4	0	4	0	4	0
B	6	0	6	2	8	2
C	5	4	9	4	9	0
D	7	4	11	5	12	1
E	3	9	12	9	12	0
F	4	12	16	12	16	0
G	8	6	14	8	16	2

Critical path: A → C → E → F. Project duration = 16 weeks. (Other paths: A–D–F = 15; B–E–F = 13; B–G = 14.)

Q8 (b)

(i) **BI software**: a set of applications that collect, integrate, analyse and present business data, such as dashboards, reports, OLAP, data mining and ad-hoc queries. It turns raw data into actionable information for decisions on cost, profitability and customers. Examples include Power BI, Tableau and Qlik.

(ii) **Data warehouse**: a subject-oriented, integrated, time-variant and non-volatile collection of data (Inmon), designed to support decision making. It brings together data from many operational systems through ETL. Data marts and OLAP cubes are built on it.
(iii) **Lean Accounting**: applies lean thinking to accounting. It removes waste from transactions, uses value-stream costing instead of standard-cost variances, reports with simple "box scores" (operational, capacity and financial measures), and uses plain-language financial statements. It focuses on customer value and continuous improvement.

Source of year tags: CMA Final SCM Scanner (Dec 2026). Each question is set on the topic and format of the tagged past paper. The figures have been reworked, and every answer has been checked.